```
!pip install pingouin factor_analyzer -q
```

```
!pip install --upgrade numpy scipy pingouin factor_analyzer
```

Show hidden output

```
import pandas as pd, numpy as np, pingouin as pg
from scipy import stats
from factor_analyzer import FactorAnalyzer, calculate_kmo, calculate_bartlett_sphericity
```

```
from google.colab import files
up = files.upload()
nama_file = list(up.keys())[0]
print("File terbaca:", nama_file)
```

Show hidden output

```
JUMLAH_KOLOM_DEMOGRAFI = 5

df = pd.read_excel(nama_file)
butir = df.iloc[:, JUMLAH_KOLOM_DEMOGRAFI:]

print("Jumlah responden :", len(df))
print("Jumlah butir     :", butir.shape[1])
print("Data kosong      :", int(butir.isna().sum().sum()))
```

Show hidden output

```
skala = {
    "Sangat Tidak Setuju (STS)": 1,
    "Tidak Setuju (TS)": 2,
    "Netral (N)": 3,
    "Setuju (S)": 4,
    "Sangat Setuju (SS)": 5,
}

num = butir.replace(skala).apply(pd.to_numeric, errors="coerce")

# Harus 0. Jika lebih dari 0, ada jawaban yang tidak terbaca.
print("Sel yang gagal dikonversi:", int(num.isna().sum().sum()))
num.head()
```

Show hidden output

```
dimensi = {}
for kolom in butir.columns:
    nama = re.match(r"^(.*?) \[", kolom).group(1)
    dimensi.setdefault(nama, []).append(kolom)

for nama, daftar in dimensi.items():
    print(f"{nama}: {len(daftar)} butir")
```

Show hidden output

```
n = len(num)
t = stats.t.ppf(0.975, n - 2)
r_tabel = t / np.sqrt((n - 2) + t**2)
print(f"n = {n} | r tabel (α=0,05; df={n-2}) = {r_tabel:.3f}")

hasil_validitas = []
for nama, daftar in dimensi.items():
    x = num[daftar]
    total = x.sum(axis=1)
    for i, kolom in enumerate(daftar, 1):
        r, p = stats.pearsonr(x[kolom], total)
        r_koreksi, _ = stats.pearsonr(x[kolom], total - x[kolom])
        hasil_validitas.append({
            "Dimensi": nama,
            "Butir": f"B{i}",
            "r hitung": round(r, 3),
            "Corrected r": round(r_koreksi, 3),
            "p-value": round(p, 4),
            "Keterangan": "Valid" if r > r_tabel else "Tidak valid",
        })
```

```
tabel_validitas = pd.DataFrame(hasil_validitas)
tabel_validitas
```

Show hidden output

```
def cronbach_alpha(x):
    k = x.shape[1]
    return k / (k - 1) * (1 - x.var(ddof=1).sum() / x.sum(axis=1).var(ddof=1))

def kategori(a):
    return ("Sangat baik" if a >= 0.90 else "Baik" if a >= 0.80
            else "Reliabel" if a >= 0.70 else "Tidak reliabel")

hasil_reliabilitas = []
alpha_dihapus = []
for nama, daftar in dimensi.items():
    x = num[daftar]
    a = cronbach_alpha(x)
    hasil_reliabilitas.append({"Dimensi": nama, "Jumlah butir": len(daftar),
                               "Cronbach's alpha": round(a, 3), "Kategori": kategori(a)})
    for i, kolom in enumerate(daftar, 1):
        if len(daftar) > 2:
            alpha_dihapus.append({"Dimensi": nama, "Butir": f"B{i}",
                                  "Alpha jika dihapus": round(cronbach_alpha(x.drop(columns=kolom)), 3)})

tabel_reliabilitas = pd.DataFrame(hasil_reliabilitas)
tabel_alpha_dihapus = pd.DataFrame(alpha_dihapus)
display(tabel_reliabilitas)
display(tabel_alpha_dihapus)
```

Show hidden output

```
gabung = tabel_validitas.merge(tabel_alpha_dihapus, on=["Dimensi", "Butir"], how="left")

with pd.ExcelWriter("hasil_uji_validitas_reliabilitas.xlsx") as w:
    gabung.to_excel(w, sheet_name="Validitas", index=False)
    tabel_reliabilitas.to_excel(w, sheet_name="Reliabilitas", index=False)

files.download("hasil_uji_validitas_reliabilitas.xlsx")
```